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For whom the reductions count: A quantile regression analysis of class size and peer effects on scholastic achievement*

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Abstract. In this paper the controversial educational topic of class size reduction is addressed. Controlling for a large number of observable characteristics and potential endogeneity in the class size variable, an educational production function is estimated using a quantile regression technique. The "conventional wisdom" that class size reduction is a viable means to increase scholastic achievement is discounted. Rather, the results point towards a far stronger peer effect through which class size reduction may play an important role. Due to heterogeneity in the newly identified peer effect, class size reduction is shown to be a potentially regressive policy measure.

JEL classification: I21

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1. Introduction

Class size reduction is perhaps the most popular educational reform advocated by politicians and policy makers. Indeed the concept is quite simple, following the "conventional wisdom" smaller classes means allocating more instruction time per pupil, which should translate into marked increases in

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academic achievement. Much policy has been based on the premise that smaller classes benefit students in the form of higher achievement.[1] Although the picture painted by the conventional wisdom seems plausible resulting in a large amount of resources devoted to its cause, little solid evidence has been found to support the claim that class size reduction is associated with significant improvements in scholastic achievement.

Among the earliest and certainly one of the most controversial studies of the effects of schooling inputs (including class size) on scholastic achievement was the infamous Coleman Report (1966). In general, the study concluded that schooling inputs have a negligible effect on scholastic achievement.[2] Amidst the widespread claims of the benefits of class size reduction and subsequent outlay of resources towards reducing class size, it is no surprise that economists have undertaken much of the recent research on the issue. A summary of the earlier economic research on the subject includes an exhausting review by Hanushek (1986) of studies analyzing the effects of various inputs in the production of public schooling. The main finding of the survey is that the effect of class size reduction (and more generally increased expenditures on education) on achievement is ambiguous, wavering from positive to negative depending on the study. It is also noted that when the effect is found to be significantly negative (i.e. reduced class sizes increase achievement), the magnitude of the effect is minute.

Obviously, the empirical evidence up to this point by and large questions the validity of the conventional wisdom put forth above. However, past studies have for the most part addressed the causal effect of class size on scholastic achievement with a traditional ordinary least squares (OLS) approach. More recently researchers have applied an instrumental variables (IV) approach using two-stage least squares (2SLS) and, less often, field experiments in order to address the issue of potential endogeneity of class size with respect to achievement.[3] Both OLS and 2SLS are designed to estimate the *mean* or average causal effect of class size on achievement. That is to say, most of the studies estimate the effects of a change in class size on the achievement of the average individual in the sample being analyzed. This provides the researcher with an estimate of how *efficient* a reduction in class size is at boosting the achievement of the average performer.[4]

The alternative quantile regression (QR) approach goes further. It con-

[1] For example, since 1996 over $3.6 billion of state funds in California have been put forth to reduce class size, particularly in the lowest grades. Florida, Indiana, Nevada, Tennessee, and Wisconsin are among 20 US states that are currently engaged in or formulating policy to reduce class sizes. On a more international level, Dutch policy-makers reached a major decision in 1997 to allocate over one billion guilders (approximately $500 million) specifically for reducing class sizes in primary education.

[2] However, the results have been challenged both methodologically and on the grounds of flawed data collection. See Hanushek (1986) for a brief summary of the Coleman Report findings.

[3] Examples of IV applications can be found in Ackerheilm (1995) and Hoxby (1998) whereas analysis of the Tennessee STAR experiment has been done by Finn and Achilles (1990) and later by Krueger (1997).

[4] Krueger (1997) deviates from this by analyzing sub samples of the STAR experimental data. He finds substantial differences in the class size effect between boys and girls, blacks and whites, and inner city and out-of-city students. More recently, Lazear (1999) has put forth a theoretical framework in which optimal class size with respect to scholastic achievement differs between students that behave well and those that do not.

cerns itself not only with the efficiency of such a reform on the average individual, but allows the researcher to estimate the marginal effect of a given class size reduction for individuals at different points in the conditional achievement distribution. This makes it possible to assess the *equity* implications resulting from changes in class size. Here the question is more precise, where the researcher asks not what the effect of a reduction in class size is on average but for whom such effects are significant and how large they might be. In this way it is possible to ascertain how the potential benefits from additional teaching resources are distributed across individuals with different levels of achievement.[5] To this end, the following study applies a QR technique to the class size issue in order to better understand for whom reductions in class size count and how large the effects are across various points of the conditional achievement distribution.

The paper is organized as follows: Section 2 briefly describes the econometric background of QR and includes a brief survey of an empirical analysis that has employed the technique in the context of educational achievement; Section 3 follows with a discussion of the potential endogeneity of class size and a description of the model used; Section 4 provides a description of the data while the empirical results are reported in Section 5; Section 6 summarizes and concludes.

2. Econometric background and previous research

QR was developed by Koenker and Basset (1978) as a robust alternative estimation technique to that of OLS when the assumption of a normally distributed error term was in question. The authors make a strong point against the excessive use of the OLS estimator resulting from the widespread carefree acceptance of a normal "Gaussian" distribution of errors. By construction, quantile parameter estimates are extremely robust to outliers. Also, robustness to monotonic transformations of the dependent variable make it quite flexible with respect to functional form.

An added merit of the estimator concerns what it intends to estimate. Rather than estimating the effects of independent variables at the conditional *mean*, the quantile estimator is designed to predict effects of explanatory variables at different points (quantiles) in the conditional distribution of the dependent variable.[6] For this reason the estimator has become very attractive to researchers interested in determining differential effects of regressors at varying points in the distribution of a dependent variable as well as assessing the underlying causes of changes in a distribution over time. Prime examples in the literature include Buchinsky (1994, 1998), where factors associated with

[5] For instance, it is not hard to imagine that, although on average there is a negligible effect of reducing class size on achievement, for certain groups (such as students who are marginal achievers or "at risk"), smaller classes may offer significant benefits.

[6] The distinction between the statistical and economic theoretic justification for using quantile over least-squares techniques is made clear by Hamermesh (1999), "... the choice among LS (least squares), LAD (least absolute deviation) and other estimators has no basis in the economic behavior one is modeling, being purely a matter of one's beliefs about the informational content of the outliers in the data. The related issue, whether underlying behavior differs at different points of the distribution of *Y*, is economic."

changes in the US wage distribution (increased wage inequality) and rates of return to education over time are analyzed; Bedard (1998) which uses the technique to address the effect of school quality on the distribution of male earnings in Canada; and, Eide and Showalter (1998) that analyzes the effect of school quality on student performance.

In the available literature, the last reference above (Eide and Showalter, 1998) provides the only study to date that uses QR to explicitly address the relationship between school quality (including class size) and scholastic achievement among high school students in the US. The duo use the "interpretive" reasoning set forth above in employing QR as an alternative to OLS.

> "... while increases in per pupil spending may not matter for average test scores, it would be useful to know if increased spending increases test scores at the bottom of the conditional distribution. In short, we not only address the question, 'does money matter?' but we also ask the question, 'for whom does money matter?'"

The analysis utilizes the High School and Beyond dataset from which they construct a cohort of public school individuals that were high school sophomores in 1980 and finished out their secondary education at the same school (i.e. did not "drop out" or transfer to a new school before graduating). The achievement proxy (dependent variable) used is the change in score on a standardized test of mathematics between an individual's sophomore and senior year. The regressions control for several characteristics specific to the individual, class, school and district.[7]

Regressions on the cohort are carried out estimating the effects of the various regressors on achievement quantiles at the 5th, 25th, 50th, 75th and 95th percentiles. In addition, OLS is used to estimate the mean effects of the educational inputs and individual characteristics. Starting with the OLS results the authors note a general insignificance among variables "typically thought to be under the control of policy makers" (i.e. pupil-teacher ratio, duration of school year, district level per pupil expenditures, and educational attainment of instructors). Total enrollment in the school is found to be significantly positive.[8] Moving to the quantile estimates, insignificant effects of schooling length are found for the two lowest quantiles while positive significant effects prevail for achievers at the median, 75th and 95th percentiles. Per pupil district-level expenditures have a significantly positive effect on the achievement of individuals at the 5th percentile. As in the case of OLS, the strongest findings are associated with the total school enrollment where the effect is highly significant and positive for all but the 95th percentile. However, although significant, the school enrollment effect is quite small.[9] Two hypotheses for this finding are put forth. The first simple interpretation considered is that there

[7] Class, school and district-specific characteristics include: pupil-teacher ratio; district level per pupil expenditures; the fraction of teachers with an advanced degree; school enrollment; and, length of school year. Individual-specific characteristics include: initial math score in sophomore year; gender, race/ethnicity, presence of both parents in household; parental educational attainment; family income and size; and, region and community of residence.

[8] This result has since been found by Angrist and Lavy (1998) and Dobbelsteen *et al* (1998) for Israeli and Dutch data, respectively.

[9] For example, an increase in school enrollment of one standard deviation (787 individuals) is, *ceteris paribus*, associated with an increase in math score gain on the order of 0.03 to 0.07 standard deviations dependent on estimated quantile.

may be increasing returns to scale with respect to school size. However, this is discarded in favor of the second "flexibility" hypothesis. Because the enrollment effect is measured holding both the pupil-teacher ratio and per pupil expenditure constant, increased enrollment simply equates to increase the number of classes in a school. This, in turn, allows a greater flexibility in the grouping of students according to ability level, whose positive effect is manifested in the school enrollment variable. Finally, the regressions present weak results with respect to instruction time per student (pupil-teacher ratio) and quality of instruction (educational attainment of teachers) which:

> "... tend to have little effect on test score gains with generally insignificant coefficients of mixed signs across the various regressions."

3. The endogeneity problem and model of achievement

Before presenting the model used in our analysis a major econometric concern in the recent class size literature should be discussed. In particular, many recent studies have addressed the possibility of class size being *endogenous* with respect to scholastic achievement. Endogeneity of this variable may significantly bias OLS estimates of the class size effect on achievement. To see this, consider the following example where an individual's class size is predetermined. This is the case where parents may choose to enroll their children in schools with smaller or larger class sizes. Suppose a systematic pattern exists such that parents with a greater interest in the scholastic success of their children invest in education outside of the classroom (for example, by having them privately tutored) in addition to placing them in schools with smaller classes. If the activites outside of school that promote learning are not explicitly controlled for in the OLS educational production function the estimated class size effect would be overstated (biased downwards). Alternatively, it is not hard to imagine a scenario in which a bias in the opposite direction might occur. Suppose a non-random pattern of enrollment emerges where parents of less able children opt to send their kids to schools with smaller classes that they might receive more attention and hence, a better education. In this case, smaller classes would be correlated with the achievement of less-abled students resulting in an overstated OLS estimate of the class size effect (biased upwards).

Given the possibility of biased class size estimates, it would seem most prudent to attempt to test and control for the endogeneity of class size. Few empirical papers to date have taken into account the possibility of endogenous regressors in a QR framework. In the study by Abadie *et al* (1998) a "quantile treatment effects" estimator (QTE) is developed to account for endogeneity of the fertility decision when analyzing the distribution of income in a quantile setting. A second work by Ribiero (2001) investigates the possible endogeneity of wages and income with respect to hours worked in a QR framework by employing a two-stage least absolute deviation estimator (2SLAD) originally developed by Amemiya (1982).[10]

[10] In addition, the reader is referred to Powell (1983) for the conditions under which QR estimates using fitted values as regressors are consistent as well as an alternative derivation of the asymptotic covariance matrix for such an estimator.

The 2SLAD procedure is essentially the quantile estimation analog of 2SLS. Suppose we are given the following simple model with two endogenous variables:

$$y = X\beta + Y\gamma + \varepsilon \tag{1}$$

$$Y = X\delta + Z\varphi + \rho, \tag{2}$$

where y is the dependent variable of primary interest; X is a matrix of exogenous variables; Y is an endogenous regressor of y (i.e. a significant correlation between Y and ε is thought to exist); Z is a matrix of valid instrumental variables; and ε and ρ are random error terms. The first stage simply consists of equation (2) being estimated using OLS. In the second stage the endogenous regressor in equation (1), Y, is replaced with fitted values calculated from the estimated first stage equation, $\hat{Y}$, and the equation estimated using QR at a given quantile of interest, τ. Slightly more complicated is the following formulation by Chen (1988) of the correct covariance matrix for a given quantile estimated by 2SLAD:

$$\sqrt{n}(\hat{\theta}_\tau - \theta_\tau) \xrightarrow{D} N(0, C_1 D^{-1}) \tag{3}$$

$$C_1 = E([f(F^{-1}(\tau))]^{-1}\varphi_\tau(v_i) - V_i\gamma)^2, \tag{4}$$

where n is the number of observations; $\hat{\theta}_\tau$ is the vector of estimated second-stage coefficients; F and f are the cumulative and density functions of v_i, the error term from the reduced form equation of y; $\varphi_\tau(v_i) = \tau - I(v_i < 0)$, where $I(E)$ is an indicator function equal to 1 when event E is true and 0, otherwise; V_i is the error term of the reduced form equation of Y (i.e. the first-stage errors); and, D^{-1} is the second-stage moment matrix *plim* $n^{-1}(W'W)$ where $W = [\hat{Y}, X]$.[11]

The approach used here is quite similar to that implemented by Angrist and Lavy (1998) who construct an instrument for class size using a rule contained in medieval Jewish law. Maimonides' Rule (named after the rabbinical scholar who scribed it) dictates the number of teachers to be allocated to a given class based on the number of participating students. The authors show the rule to produce a functional relationship between average class size (as predicted by the rule) and enrollment that is *non-monotonic* and *discontinuous*, thereby giving support to the validity of the former as an instrument for actual average class size.[12]

The instrument used here is based on a similar rule applied by the Dutch Ministry of Education which links total school enrollment (weighted by socio-economic status of the student body) to the number of teachers a school receives funding for.[13] In order to determine the number of instructors a school

[11] Note, the expectation $E(.)$ can be estimated by its sample analog $\Sigma(.)^2/n$. The error terms are related by the following identity $\varepsilon = v - V\beta$.

[12] Support for the validity of such an instrument can be found in the literature on regression discontinuity design (see Campbell and Stanley, 1963).

[13] Note that Angrist and Lavy's instrument is based on enrollment at the grade level, whereas here the instrument is based on total enrollment at the school level.

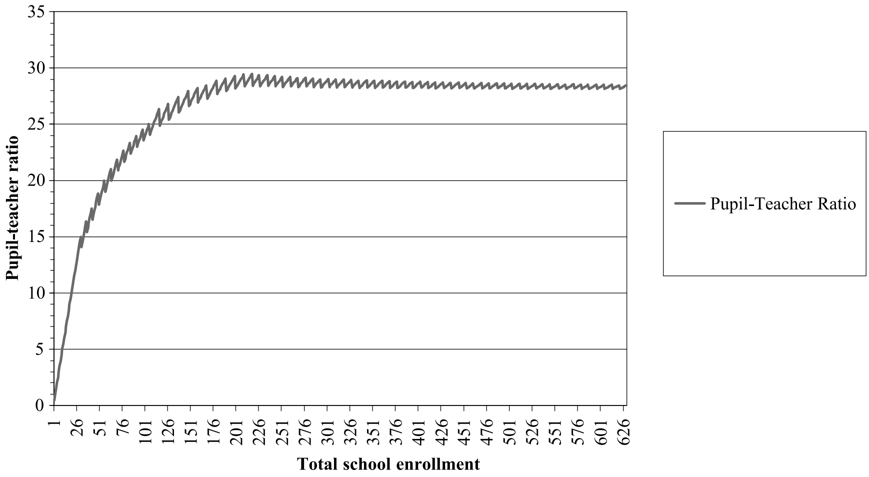

Fig. 1. Graph of Pupil-Teacher Ratio (as determined by enrollment function)

is allocated, the weighted number of students (WNS) of a school must first be calculated using the following formula:

$$WNS = INT\left(1.03 * \max\left\{INT\left(\sum_{i=1}^{N} w_i - 0.09 * N\right), N\right\}\right), \tag{5}$$

where N is total enrollment in the beginning of a school year and w_i is a weight factor denoting the socio-economic status (SES) of pupil i. The number of teachers that a school is able to appoint can then be looked up in tables mapping the WNS to full-time teaching units. For example, a school with a number of weighted students up to 30 can appoint two full-time teachers, and the number of teachers increases by an alternating 0.2 or 0.3 full-time units for every 6 to 11 additional weighted students.

For each school, known is the number of pupils and their various weight factors (see Section 4, below), making it possible to derive the number of teachers that can be appointed by the school based on the ministerial rules. The instrument used then is the predicted average class size, calculated as total school enrollment divided by the predicted number of allocated teachers using the ministerial rules. Henceforth, I refer to this instrument as the "Enrollment Function Determined" class size, or EFD class size for short. Figure 1 provides a sketch of the average EFD class size as a function of enrollment.

From the characteristic "saw tooth" shape of the figure the function is clearly non-monotonic and discontinuous and thus, may serve as an instrument for class size. With our constructed instrument at hand the estimated model corrected for endogeneity then becomes:

$$y_{ics} = X_i'\alpha + Z_c'\lambda + W_s'\chi + n_{ics}\mu + \varepsilon_{ics} \tag{6}$$

$$n_{ics} = X_i'\beta + Z_c'\gamma + W_s'\delta + EFD_s\nu + \rho_{ics}, \tag{7}$$

where y_{ics} is the percentile score (in arithmetic or language) of pupil i in class c

of school s; X_i, Z_c and W_s are vectors of variables pertaining to the individual, the class and the school, respectively; n_{ics} is the number of pupils in pupil i's class c within school s; EFD_s is the enrollment determined average class size of school s; and ρ_{ics} and ε_{ics} are individual-specific error terms corresponding to the achievement and class size equations, respectively.

4. Description of data

The following analysis makes use of PRIMA, a longitudinal survey containing information on Dutch pupils who were enrolled in grades 2, 4, 6 and 8 in the 1994/1995 school year.[14] Questionnaires were administered to teachers, parents, and headmasters of approximately 800 primary schools throughout the Netherlands. Of these, 400 were chosen to form a nationally representative sample of "regular" schools used in this analysis. Here, the sample is limited to only the observations on individuals from grades 4, 6 and 8, as the information on class size for grade 2 is too unreliable.[15]

The PRIMA survey provides a wealth of information at the pupil, class, and school levels. Individual grade-level percentile rankings on standardized arithmetic and language tests serve as proxies of scholastic performance and are therefore used as dependent variables that are regressed on various characteristics specific to pupils, classes and schools. At the individual level indicators of gender and socio-economic status (SES) are included as covariates. The SES measure takes the form of four indicator variables denoting weight factors that are a function of a pupil's social background.[16] The weight factor ranges from 1.00 to 1.90 and is defined as follows:

Weight factor value	Definition of weight factor category
1.90	Pupils with foreign-born parents that satisfy one of the following conditions: 1) father, mother or guardian has at most a VBO-level education;[17] 2) the primary earning parent or guardian has either a job involving physical labor, or has no income from labor at the time of interview.
1.70	Pupils whose parents are transients.
1.40	Pupils living in a boarding school or a foster home, and whose parents are masters of a ship.
1.25	Pupils of whom either parents or guardians have at most an education at VBO-level.
1.00	Other pupils (reference group).

[14] This data has previously been used in studies by Bosker and Hox (1996) and Dobbelsteen *et al* (1998).

[15] The documentation of the PRIMA survey states that a considerable number of teachers in grade 2 seem to have given erroneous information on class size. Probably they filled in the total number of pupils in the whole year group of that grade level instead of the number in their own class.

[16] Note, that SES is used to determine the amount of resources a school is given for a certain pupil; schools are allocated more resources for pupils from disadvantaged backgrounds (SES > 1) than for pupils with a "normal" (SES $= 1$) background.

[17] VBO stands for Voorbereidend Beroepsonderwijs or secondary vocational education, which serves as the lowest form of secondary schooling in the Dutch educational system.

At the class level the regressions control for the following: class size, teacher's gender and years of experience, class averages of pupil gender and weight factor, dual teacher class (whether more than one teacher has taught the class), and multi-grade class (whether the class combines individuals from more than one grade level).[18] School level controls include indicators denoting the denomination of the school, the average SES within a school, and total enrollment of the school.[19] The sample has been limited to only those observations without missing values for the variables used in the analysis.[20]

Appendix A contains descriptive statistics by grade level for the variables used in the analysis. A brief look at the table reveals nothing too surprising. Class sizes are remarkably similar across grades; the average 4th grade class has between 25 and 26 students and increases to 26 and 27 for grades 6 and 8, respectively. Over half of all students are not considered disadvantaged as measured by the SES weighting scheme (i.e. their SES weight factors equal 1). Another 35 to 40 percent are slightly disadvantaged with an SES weight of 1.25. What is perhaps most noteworthy concerns the teacher characteristics. In particular, the proportion of instructors that are female drops sharply as grade level increases; whereas over three quarters of the 4th grade teachers are women, this fraction drops to under one half in 6th grade and just over one quarter in 8th grade. In addition, the average 8th grade instructor has over one year more experience than their counterparts teaching grades 4 and 6.

5. Empirical analysis

A preliminary look at the relationship between class size and achievement across the five "common" quantiles to be estimated (the 10th, 25th, 50th, 75th, and 90th percentiles) is provided in Figures 2a and 2b. Each figure contains, for each grade level, plots of mean achievement (as proxied by test scores) versus class size category by quantile.[21] Beginning with Figure 2a, for 4th graders a very weak upward trend in the mathematics achievement is found for all five quantiles. Note that the 10th, 25th and 50th percentiles are just about parallel throughout the range of class size categories denoting that there is most likely no significant differential in class size effect across these quantiles. For 6th graders, there seems to be a slight dip in math achievement across all quantiles, save the 25th, associated with a one-category increase in class size (from 5–14 to 15–19 students). This drop is most prominent for those in the 90th percentile. However, an additional increase in class size is met with increases in achievement across the upper four quantiles. Further increases have only very slight effects on math achievement producing remarkably flat class size/achievement profiles across the various quantiles. The 8th

[18] Note that all independent variables denote lagged indicators of individual, class and school characteristics the year prior to the survey (1993–94), whereas achievement measures are proxied by scores on tests taken within the first few months of the survey year (1994–95).

[19] The three categories of religion/ideology are: Protestant, Roman Catholic and "Other" which include Islamic, general special and other ideologies. Public schools serve as the reference group.

[20] The total number of observations contained in the PRIMA reference sample are 5,698, 5,368 and 5,608 for the 4th, 6th and 8th grades, respectively. The number dropped due to missing values are 1,245, 1,278 and 1,364 leaving a total of 4,453, 4,090 and 4,244 usable observations respectively for 4th, 6th and 8th grade.

[21] Class size categories are as follows: 5–14, 15–19, 20–24, 25–29, 30–34, and 35 and greater.

grade profiles show an even flatter relationship between class size and math achievement. Most notable is an increase in math score for the lowest three quantiles associated with a move from class sizes of 20–24 to those containing 25 to 30 students. The graphs in Figure 2b exhibit quite similar profiles for language achievement. In brief, the data shows relatively flat profiles except for the familiar "dip" in achievement associated with 6^{th} grade class sizes of 15–19 individuals.[22]

Clearly, there seems to be no consistent pattern that emerges with respect to achievement and class size. Two points should be made clear. First, analysis of the raw distributions provides little evidence in favor of the conventional wisdom, which purports a general negative relationship between class size and scholastic achievement. Examination of the raw achievement distribution shows both increases and decreases in achievement associated with increases in class size. Second, the quantile achievement profiles give no strong indication of divergence or convergence with respect to class size. This suggests that there is likely no discernable difference in class size effect between different quantiles of the conditional achievement distribution. However, one must be wary of drawing conclusions from such a "rough" sketch of the relationship between class size and achievement. A more in-depth analysis that controls for several observable characteristics is necessary to properly ascertain whether there is a significant effect of class size on scholastic achievement and whether any such effect is heterogeneous across the achievement distribution.

Estimations of equations (6) and (7) using 2SLS and 2SLAD for the five common quantiles have been calculated. Table 1a contains the results of the math achievement regressions by grade level. As we are primarily concerned with the effect of class size on achievement, all additional coefficients except enrollment have been omitted from the table. The estimated parameters are reported with robust standard errors in parentheses.[23] The first column for each set of regressions contains estimated mean class size effects using 2SLS for 4^{th}, 6^{th} and 8^{th} graders, respectively. The point estimates imply there is a significant *positive* effect of class size on math achievement for the average pupil in 8^{th} grade. The mean effect of school enrollment proves to be insignificant at conventional levels.

Turning our attention to 2SLAD estimates, significant findings are only found for those in the 8^{th} grade. Here, similar significant positive class size coefficients are found for all but the extreme quantiles. More simply, an 8^{th} grade individual at the 25^{th}, 50^{th} and 75^{th} percentiles of the conditional distribution of math achievement is expected to significantly *benefit* from an increase in class size. The magnitude of this effect is largest and highly significant (at the 1%-level) at the median.[24] Quantile estimates of the school

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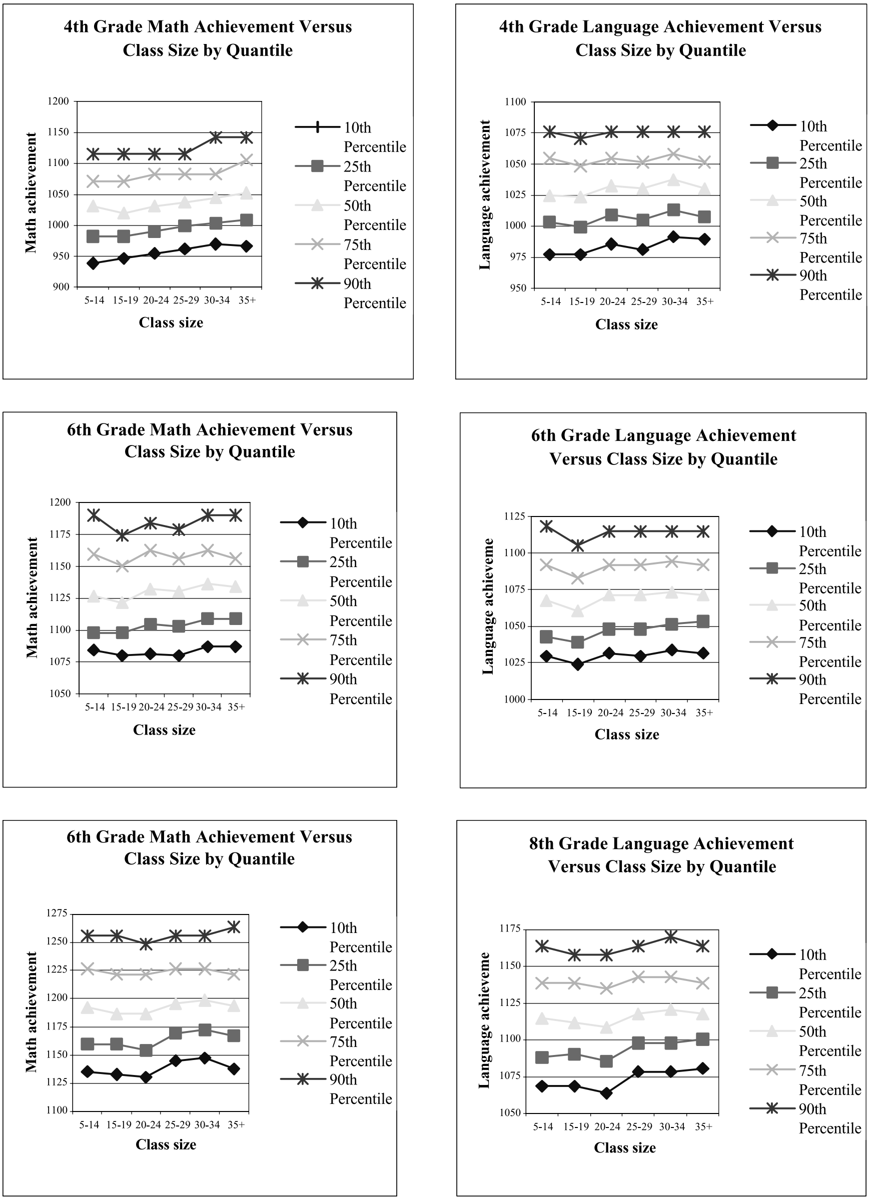
[22] Further investigation shows that this dip can be pinpointed to 6^{th} grade females who experience a marked drop in both math and language achievement across all quantiles associated with an increase in class size from 5–14 to 15–19 students.

[23] Robust standard errors for the OLS regressions are calculated using Roger's (1993) generalization of the estimator put forth by Huber (1967), which takes into account possible correlations between the errors of observations *within* groups (in our context schools). Standard errors for the estimates using QR have been calculated using the method from Chen (1988), described above in Section 3.

[24] Although the class size estimate is significant, the size of the benefit is relatively small; an increase in class size of 10 students (1.64 standard deviations) is only expected to increase math achievement of such an individual by 15.16 percentile points (0.54 standard deviations).



a b

Fig. 2a.
4th Grade Math Achievement Versus Class Size by Quantile
6th Grade Math Achievement Versus Class Size by Quantile
8th Grade Math Achievement Versus Class Size by Quantile

Fig. 2b.
4th Grade Language Achievement Versus Class Size by Quantile
6th Grade Language Achievement Versus Class Size by Quantile
8th Grade Language Achievement Versus Class Size by Quantile

Table 1a. Effect of class size on math achievement of 4th, 6th and 8th graders (two stage least squares and two stage least absolute deviation estimates)

	4th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.348 (0.362)	0.065 (0.674)	0.612 (0.582)	0.217 (0.473)	0.193 (0.428)	0.591 (0.463)
Enrollment	0.015 (0.013)	0.016 (0.019)	0.018 (0.016)	0.031 (0.013)**	0.011 (0.012)	−0.007 (0.013)
(Pseudo) R-squared	0.0863	0.0353	0.0543	0.0553	0.0005	0.0343
Hausman test	−0.239 (0.401)					
Partial R-squared	0.3839***					
Number of observations	4,453	4,453	4,453	4,453	4,453	4,453
	6th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.223 (0.273)	0.171 (0.506)	0.395 (0.543)	0.446 (0.385)	0.118 (0.446)	0.182 (0.493)
Enrollment	−0.001 (0.010)	−0.003 (0.017)	−0.001 (0.018)	0.003 (0.013)	−0.006 (0.015)	−0.006 (0.016)
(Pseudo) R-squared	0.0826	0.0352	0.0509	0.0596	0.0495	0.0268
Hausman test	−0.401 (0.300)					
Partial R-squared	0.4671***					
Number of observations	4,090	4,090	4,090	4,090	4,090	4,090

Table 1a. (cont.)

	8th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.857 (0.367)**	0.814 (0.553)	1.050 (0.414)**	1.516 (0.532)***	0.991 (0.500)**	−0.025 (0.429)
Enrollment	−0.012 (0.010)	−0.006 (0.019)	−0.013 (0.014)	−0.029 (0.018)	−0.018 (0.017)	0.004 (0.015)
(Pseudo) R-squared	0.1135	0.0723	0.0934	0.0931	0.0674	0.0404
Hausman test	−0.974 (0.370)***					
Partial R-squared	0.4310***					
Number of observations	4,244	4,244	4,244	4,244	4,244	4,244

Table 1b. Effect of class size on language achievement of 4th, 6th and 8th graders (two stage least squares and two stage least absolute deviation estimates)

	4th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.137 (0.371)	0.456 (0.668)	0.449 (0.566)	0.094 (0.398)	0.113 (0.390)	−0.227 (0.427)
Enrollment	0.010 (0.014)	0.012 (0.018)	0.009 (0.016)	0.016 (0.011)	0.003 (0.011)	−0.001 (0.012)
(Pseudo) R-squared	0.1435	0.0568	0.0919	0.0999	0.0792	0.0493
Hausman test	−0.270 (0.414)					
Partial R-squared	0.3839***					
Number of observations	4,453	4,453	4,453	4,453	4,453	4,453

Table 1b. (cont.)

	6th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	−0.261 (0.270)	−0.249 (0.567)	−0.245 (0.424)	−0.116 (0.482)	−0.431 (0.435)	−0.255 (0.436)
Enrollment	0.014 (0.011)	0.020 (0.019)	0.025 (0.014)*	0.007 (0.016)	0.008 (0.014)	0.008 (0.014)
(Pseudo) R-squared	0.1222	0.0432	0.0739	0.0847	0.0690	0.0391
Hausman test	0.070 (0.335)					
Partial R-squared	0.4671***					
Number of observations	4,090	4,090	4,090	4,090	4,090	4,090
	8th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.519 (0.284)*	0.626 (0.563)	0.841 (0.468)*	0.498 (0.372)	0.367 (0.324)	0.192 (0.423)
Enrollment	−0.004 (0.009)	−0.003 (0.020)	−0.011 (0.016)	0.000 (0.013)	−0.001 (0.011)	−0.008 (0.015)
(Pseudo) R-squared	0.1523	0.0766	0.0992	0.1005	0.0909	0.0686
Hausman test	−0.559 (0.293)*					
Partial R-squared	0.4310***					
Number of observations	4,244	4,244	4,244	4,244	4,244	4,244

Note: Dependent variables are percentile ranking of pupils' scores on standardized arithmetic or language tests. All regressions also include additional controls for: a constant, four individual SES-class dummies, a dummy for pupil's gender, percentage of females in pupil's class, class average SES, teacher's gender and experience, dummy for dual teacher class, dummy for multi-grade class, three dummy variables for school's denomination and total school enrollment. Robust standard errors taking into account grouped heteroskedasticity for least squares equations and using design matrix bootstrapped for quantile regressions are reported in brackets. ***/**/* indicate significance at the 1%/5%/10%-levels. Pseudo R-squared equals $\left(1 - \frac{\text{sum of weighted deviations about estimated quantile}}{\text{sum of weighted deviations around raw quantile}}\right)$.

enrollment effect yield but one (positive) coefficient that is significant at conventional levels. Hausman tests reveal significant endogeneity of class size with respect to the math achievement of those in 8^{th} grade. The negative sign of the coefficient shows the direction of the bias on the uncorrected OLS estimate. This implies there is some sorting mechanism by which 8^{th} grade pupils with above average ability are assigned to smaller classes and/or those with below average ability are enrolled in classes with a larger number of students.

The regressions for language achievement are included on the following table. Here, we find a marginally significant positive class size effect for 8^{th} grade using 2SLS and at the 25^{th} percentile using 2SLAD. Again the Hausman test produces a negative coefficient implying the existence of non-random selection of more (less) able students in to smaller (larger) classes. Precision set aside, only six of the fifteen 2SLAD point estimates for class size produce a negative sign consistent with the conventional wisdom. The enrollment variable yields only one marginally significant (positive) coefficient for 6^{th} graders at the 25^{th} percentile.

The results so far coincide with the majority of past findings on class size and achievement. That is, no consistent evidence is found in support of the conventional wisdom that class size reductions are beneficial to scholastic achievement.[25] To the contrary, some of the resulting estimates for mathematics suggested just the opposite, that *increases* in class size may in fact be beneficial to scholastic achievement! The question must then be asked, has the relationship between class size and scholastic achievement been modeled correctly?

Although the evidence thus far suggests there is no clear difference in achievement between those in larger and smaller classes, we should not prematurely discount the hypothesis put forth by the conventional wisdom. Obviously, class size alone is not the sole determinant of scholastic achievement and, although the previous regressions control for a great many additional factors that are no doubt of importance to educational production, perhaps there is another piece to the "puzzle" that is still missing. More precisely, if in the previous regressions there was some unobserved factor that proved to be positively correlated with both class size and achievement then it very well might be the case that the former "picked up" part of the effect of the omitted variable. Needless to say, this would undoubtedly obscure the estimated class size effect.

One aspect of educational production that has been largely neglected in the economics literature while receiving much attention from educational researchers concerns the interaction of those learning within the educational environment (i.e. the existence of a "peer effect"). The idea that individuals learn not only from the instructor but also from their fellow students is well documented. Social cognitive learning theory emphasizes that people acquire knowledge, skills and attitudes by observing others (Bandura, 1986 and Schunk, 1991). In addition, educational practices such as the use of peer

[25] Hanushek (1986), in his survey article on educational production, finds of 136 separate estimates of the class size effect that 13% were significantly positive and 20% significantly negative. Furthermore, the insignificant positive and negative estimates accounted for 25% and 31% of the 136 estimates surveyed, respectively. 23% of the estimates produced estimates with what is referred to as an "unknown" sign.

models and ability grouping are based on the idea that pupils learn by observing and working together with other similar individuals. In this literature evidence can be found of the positive effects of both ability grouping (Slavin, 1987) and peer modeling (Schunk *et al*, 1987) on the school performance of pupils.

Research on peer modeling suggests that both the competencies of fellow peers and number of role models can affect a pupil's performance. Similarity in competence appears to be particularly important for purposes of self-evaluation. It is found that observing the success of individuals similar to oneself raises the observer's self-efficacy and motivates her or him to perform the same tasks.[26] This is especially true among children who have experienced difficulties learning skills (Schunk and Hanson, 1985). Moreover, both Schunk *et al* (1987) and Bandura and Menlove (1968) find advantages for multiple peers, presumably as a larger number of peers increases the probability that observers will perceive themselves as similar in competence to at least one of the role models. Because similarity in competence appears to be important for purposes of self-evaluation, there may exist a potential "indirect" effect resulting from changes in class size by which a resulting decrease (increase) in the expected number of "similar" students via smaller (larger) class sizes could account for the failure of the previous results to support the conventional wisdom.

The estimated non-negative (and positive) effects on achievement due to the presence of a larger number of similar individuals may occur for reasons other than pupil interaction. One possible mechanism leading to a positive relation between performance and the presence of "similar" classmates is somewhat analogous to the conventional wisdom. The idea is that teachers may target their instruction to groups of children with similar competence levels. If this is the case and instruction time is divided in proportion to the size of the group, members of larger groups should receive more instruction. Therefore, larger classes of fewer similar groups should provide a positive correlation between class size and achievement.[27] Alternatively speaking, if group targeting serves as a dominant teaching method and smaller classes tend to be less homogeneous with respect to ability, a similar positive relationship between class size and achievement should emerge.[28] However, regardless of the exact mechanism driving the non-negative correlation between class size and achievement, it seems likely that peer effects could at least partially explain this phenomenon.

Dobbelsteen *et al* (1998) address this issue and find that accounting for the peer effect produces class size estimates that are more in line with the conventional wisdom (i.e. achievement becomes negatively correlated with class size). In the study controlling for the number of similar peers in an in-

[26] France-Kaatrude & Smith (1985) show that children that were allowed to compare their performances with a similarly performing peer compared themselves more often and demonstrated greater task persistence than children that were offered comparisons with superior or inferior peers.

[27] An alternative hypothesis is that the presence of more similar students enhances the level competition in the learning environment spurring individuals to work harder.

[28] Unfortunately, the data at hand does not provide a detailed account of the degree to which a teacher performs group instruction and how his/her time is allocated with respect to "similar" groups.

dividual's class is achieved via the inclusion of the variable "number of classmates with similar IQ".[29] In addition to producing estimates of the class size effect that are more in line with the conventional wisdom, the extra control variable proved to have a strongly significant positive effect on scholastic achievement. In our context the important question is to discern whether the resulting estimates calculated by Dobbelsteen *et al* are applicable across the whole conditional achievement distribution or if they are only indicative of the class size and peer effects at certain points of the distribution.

Tables 2a and 2b presents estimation results from OLS and quantile regressions with specifications identical to those reported in Tables 1a and 1b except that the peer effect control variable described above, "number of classmates with similar IQ", is now included. The correlation coefficient between this variable and actual class size equals 0.45 in grade 4, 0.33 in grade 6, and 0.32 in grade 8.[30] Looking at the math results we find now that all of the 2SLS class size estimates have dropped considerably with the 6th grade point estimate becoming negative. The corresponding peer effects prove to be positive and highly significant across the board. For each similar peer an individual learns with, he or she is expected (on average) to rank 0.61 to 0.97 percentage points higher on the math exam. Shifting our attention to the quantile regressions we can see how these potential effects are distributed over the achievement distribution. Despite a general drop in the class size effect at every quantile across all three grades, only three of the resulting point estimates are significant (at each the 1, 5, and 10%-levels) and remain positive. Far more interesting are the accompanying peer effects, whose point estimates dominate those of class size in both magnitude and significance. A quick tally shows that seven of the point estimates are highly significant, three are significant at the 5%-level, while the remaining five do not differ significantly from zero. Turning to the enrollment effects we find that all the 2SLS estimates retain their original signs and remain insignificant. In addition, enrollment now has a marginally significant negative effect only for 8th graders at the median.

The results for language achievement on Table 2b are quite similar. In brief, while all three of the estimated mean effects of class size have decreased in magnitude, only that of the 6th grade significantly differs from zero (at the 10%-level). Similarly, although most of the 2SLAD class size coefficients dropped with two turning negative, all are insignificant. The peer effect on language achievement proves significant in six of the fifteen quantile estimates however the size of the effects is not as pronounced as for math. Finally, the enrollment effect is insignificant at all points across the achievement distribution.

Perhaps the most striking finding resulting from this alternative specification is the pattern of the newly incorporated peer effect with respect to the conditional achievement distributions. In particular, in four of the six grade-specific sets of quantile regressions, "number of classmates with similar IQ" is

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[29] Classmates are defined as having a similar level of competence to that of the given individual if they fall within a range of plus or minus half a grade-level standard deviation around the given pupil's IQ.

[30] It should also be noted that the number of similar pupils is not a direct measure of class homogeneity with respect to IQ such as variance. Although the number of pupils in the class with similar IQ and the within-class variance of IQ are negatively correlated, this correlation is relatively weak: −0.25 in grade 4, −0.20 in grade 6, and −0.17 in grade 8.

Table 2a. Effect of class size on math achievement of 4th, 6th and 8th graders controlled for peer effect (two stage least squares and two stage least absolute deviation estimates)

	4th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.084 (0.411)	−0.311 (0.674)	0.154 (0.611)	0.088 (0.513)	0.156 (0.491)	0.386 (0.609)
Enrollment	0.008 (0.013)	0.007 (0.016)	0.008 (0.014)	0.019 (0.012)	0.010 (0.011)	−0.004 (0.014)
Peer effect	0.969 (0.240)***	1.605 (0.439)***	1.424 (0.398)***	1.185 (0.334)***	0.319 (0.320)	0.086 (0.397)
(Pseudo) R-squared	0.1052	0.0626	0.0738	0.0654	0.0520	0.0355
Hausman test	−0.256 (0.454)					
Partial R-squared	0.3612***					
Number of observations	4,453	4,453	4,453	4,453	4,453	4,453
	6th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	−0.033 (0.280)	−0.028 (0.474)	0.068 (0.543)	0.004 (0.498)	0.047 (0.440)	0.152 (0.550)
Enrollment	−0.002 (0.009)	−0.003 (0.015)	−0.006 (0.017)	0.007 (0.015)	−0.008 (0.014)	−0.008 (0.017)
Peer effect	0.953 (0.167)***	1.018 (0.314)***	1.514 (0.359)***	1.238 (0.329)***	0.618 (0.291)**	0.106 (0.364)
(Pseudo) R-squared	0.0999	0.0464	0.0664	0.0674	0.0535	0.0272
Hausman test	−0.284 (0.305)					
Partial R-squared	0.4467***					
Number of observations	4,090	4,090	4,090	4,090	4,090	4,090

Table 2a. (cont.)

	8th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.690 (0.397)*	0.598 (0.508)	0.926 (0.436)**	1.262 (0.479)***	0.855 (0.486)*	−0.028 (0.495)
Enrollment	−0.012 (0.010)	−0.013 (0.016)	−0.013 (0.014)	−0.025 (0.015)*	−0.014 (0.015)	0.004 (0.016)
Peer effect	0.608 (0.197)***	0.810 (0.335)**	0.870 (0.288)***	0.658 (0.317)**	0.270 (0.321)	0.118 (0.327)
(Pseudo) R-squared	0.1279	0.0863	0.1051	0.1017	0.0702	0.0407
Hausman test	−0.969 (0.394)**					
Partial R-squared	0.4070***					
Number of observations	4,244	4,244	4,244	4,244	4,244	4,244

Table 2b. Effect of class size on language achievement of 4th, 6th and 8th graders controlled for peer effect (two stage least squares and two stage least absolute deviation estimates)

	4th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.020 (0.435)	0.387 (0.725)	0.369 (0.443)	−0.128 (0.508)	−0.022 (0.487)	−0.244 (0.503)
Enrollment	0.007 (0.013)	0.002 (0.017)	0.010 (0.010)	0.015 (0.012)	0.003 (0.011)	−0.003 (0.012)
Peer effect	0.430 (0.316)	0.464 (0.472)	0.416 (0.288)	0.548 (0.331)*	0.417 (0.318)	0.235 (0.327)
(Pseudo) R-squared	0.1483	0.0613	0.0949	0.1017	0.0809	0.0496
Hausman test	−0.298 (0.473)					
Partial R-squared	0.3612***					
Number of observations	4,453	4,453	4,453	4,453	4,453	4,453

Table 2b. (cont.)

	6th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	−0.489 (0.283)*	−0.304 (0.674)	−0.547 (0.488)	−0.560 (0.457)	−0.496 (0.469)	−0.354 (0.461)
Enrollment	0.013 (0.011)	0.017 (0.021)	0.018 (0.015)	0.008 (0.014)	0.009 (0.014)	0.007 (0.014)
Peer effect	0.848 (0.160)***	0.896 (0.446)**	1.286 (0.323)***	1.147 (0.302)***	0.413 (0.310)	0.246 (0.305)
(Pseudo) R-squared	0.1316	0.0514	0.0844	0.0918	0.0697	0.0395
Hausman test	0.186 (0.340)					
Partial R-squared	0.4467***					
Number of observations	4,090	4,090	4,090	4,090	4,090	4,090
	8th Grade					
		Quantile				
	2SLS	0.10	0.25	0.50	0.75	0.90
Class size	0.374 (0.304)	0.237 (0.626)	0.690 (0.444)	0.139 (0.410)	0.382 (0.387)	0.243 (0.444)
Enrollment	−0.004 (0.009)	−0.001 (0.020)	−0.008 (0.014)	−0.001 (0.013)	−0.006 (0.012)	−0.009 (0.014)
Peer effect	0.527 (0.154)***	0.432 (0.414)	0.730 (0.294)**	0.968 (0.271)***	0.239 (0.256)	−0.027 (0.293)
(Pseudo) R-squared	0.1612	0.0799	0.1066	0.1071	0.0920	0.0686
Hausman test	−0.539 (0.309)*					
Partial R-squared	0.4070***					
Number of observations	4,244	4,244	4,244	4,244	4,244	4,244

Note: Dependent variables are percentile ranking of pupils' scores on standardized arithmetic and language tests. All regressions also include additional controls for: a constant, four individual SES-class dummies, a dummy for pupil's gender, percentage of females in pupil's class, class average SES, teacher's gender and experience, dummy for dual teacher class, dummy for multi-grade class, three dummy variables for school's denomination and total school enrollment. Robust standard errors taking into account grouped heteroskedasticity for least squares equations and using design matrix bootstrapped for quantile regressions are reported in brackets. ***/**/* indicate significance at the 1%/5%/10%-levels. Pseudo R-squared equals $\left(1 - \frac{\text{sum of weighted deviations about estimated quantile}}{\text{sum of weighted deviations around raw quantile}}\right)$.

found to have the largest effect on the achievement of individuals at the lowest two quantiles and experiences a considerable monotonic decrease as one goes up the conditional achievement distribution.[31] Furthermore, in each set of estimations the effect becomes indistinguishable from zero by the 90th percentile. It should be noted that the decrease in size is by no means trivial as the ratio of the peer effect at the 25th to 90th percentiles averages 10.86 across the four sets of estimates. A simple interpretation for this finding is that students towards the lower end of the conditional achievement distribution are "dependent" learners, relying more heavily on the number of similar peers relative to their higher achieving counterparts that learn more "independently" (i.e. for whom the effect of similar peers is negligible).

The validity of the picture "painted" above relies on the fact that there indeed exist statistical differences across the various quantiles in math and language achievement *conditional* on the regressors used in our model. Simply put, we are interested in whether estimated class size and peer effect coefficients differ significantly across the various quantiles. Not surprisingly, the tests revealed negligible differences between quantiles with respect to class size effects while significant differences were most pervasive in the case of peer effects.[32]

After careful inspection of the results stemming from controlling for the number of classmates with similar IQ, two points deserve mention. First, the significance of the peer effect dwarfs that of the corresponding class size effect. This suggests that class composition with respect to cognitive ability is more closely related to scholastic achievement than size of class (at least within the grades and class size ranges given in our sample). Second, the results show a strong downward trend in the peer effect as one moves up the achievement distribution. As mentioned above, this implies that individuals in the lower part of the achievement distribution benefit more from learning with classmates of similar ability than those in the upper portion of the distribution. The tendency for the peer effect to diminish with higher achievement coupled with the generally insignificant class size effect therefore gives rise to the possibility of inequitable educational outcomes with no significant efficiency gains resulting from a class size reduction!

6. Conclusion

The proposition to reduce class size as a way to improve learning has pervaded debates on educational reform in recent years. Although the conventional wisdom seems logical enough, scholastic achievement should rise through the allocation of a greater amount of teaching resources per pupil, there has been little definitive evidence as to the efficacy of class size reduction in enhancing performance. In spite of the hundreds of studies that have been concerned with the class size issue, virtually all have used econometric meth-

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[31] Note, this pattern is most widespread in the case of 4th grade math where the largest peer effect is found at the lowest quantile and monotonically decreases throughout the entire conditional achievement distribution (i.e. through the 90th percentile).

[32] Similar tests have been performed in Mata and Machado (1996). In preliminary work using a standard QR technique only three of the sixty pair wise *t*-tests rejected the null hypothesis of equality of estimated class size effects across the specified quantiles. In contrast, almost two-thirds (39 out of 60) of the pair wise *t*-tests of equality of the peer effect between quantiles were rejected at the 5%-level or above.

ods that are designed to estimate the average causal effect of class size on individual achievement. While this methodology addresses the effectiveness of class size reduction in terms of average efficiency, it has little to say about the equity effects of such a policy. To this end, the QR approach of 2SLAD has been used to estimate the causal effect of class size on scholastic achievement across various points in the conditional distributions of math and language achievement of Dutch primary school students.

The initial findings on the effect of class size on math and language achievement provide little support for the conventional wisdom. The resulting point estimates of class size effect are rarely significant, changing from positive and negative depending on grade level, quantile being estimated, and dimension of achievement (math or language). A more interesting finding is evidence of significant non-random selection of more (less) able 8^{th} grade students into smaller (larger) classes.

Before completely discounting the common wisdom an alternative regression specification is considered. Drawing from social cognitive learning theory, peer effects are incorporated in the model by controlling estimations of individual acheivement for the number of classmates with similar ability. The more sophisticated specification yields decreased point estimates of the class size effect. However, imprecision renders most of the resulting coefficients insignificant at conventional levels. Much stronger in significance and magnitude is the newly incorporated peer effect which proves to be positive and dominates the class size effect across a majority of the regressions. A definitive trend emerges with respect to the estimated peer effect on math and language achievement. For all grade levels there is a significant monotonic decrease in the estimated peer effect as one goes up the conditional distribution of math achievement. This result also holds as one climbs the 6^{th} grade conditional language distribution. This suggests that individuals in the lower portion of the achievement distribution benefit more from being placed in classes with individuals of similar ablilty.

In conclusion, this study finds little support for the conventional wisdom that reducing class size improves learning "directly" via an increased allocation of teaching resources per student. In contrast, the incorporation of peer effects in the model has a profound impact on the results. It is shown that the predominant positive effect of similar peers on learning is significantly larger for those in the lower portion of the achievement distribution. Thereby class size reduction also imposes an "indirect" effect which may have drastic equity implications depending on the resulting ability composition of a class. Should reductions in class sizes consist of "thinning out" the average number of low achievers in each class, the findings imply that the resulting outcome would be inequitable as those towards the bottom of the acheivement distributions would be made worse off. In turn, rather than class size reduction, the results provide support for an alternative policy of ability grouping as a more viable means to boost scholastic achievement.

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Appendix A. Descriptive statistics of selected variables

	Grade 4			
Variable	Mean	Standard deviation	Minimum	Maximum
Math score (absolute)	1041.96	67.66	822.70	1193.40
Math score (percentile)	52.26	27.66	1.00	95.00
Language score (absolute)	1030.58	36.64	841.80	1143.30
Language score (percentile)	54.68	27.80	1.00	100.00
Class size	25.45	5.84	5.00	39.00
Number of students with similar IQ	5.99	4.32	0.00	32.00
School enrollment	216.22	103.08	23.00	552.00
Weight factor = 1.00	0.55	0.50	0.00	1.00
Weight factor = 1.25	0.35	0.48	0.00	1.00
Weight factor = 1.40	0.00	0.06	0.00	1.00
Weight factor = 1.70	0.00	0.06	0.00	1.00
Weight factor = 1.90	0.10	0.30	0.00	1.00
Gender of student (1 = Female)	0.49	0.50	0.00	1.00
Percent of class female	0.50	0.12	0.00	1.00
Class average weight factor	1.18	0.17	1.00	1.90
Teacher experience	17.68	8.14	1.00	37.00
Dual teacher class	0.38	0.49	0.00	1.00
Gender of teacher (1 = Female)	0.84	0.37	0.00	1.00
Multi-grade class	0.27	0.45	0.00	1.00
Public	0.27	0.45	0.00	1.00
Protestant	0.23	0.42	0.00	1.00
Roman Catholic	0.43	0.50	0.00	1.00
Other	0.07	0.25	0.00	1.00
School average weight factor	1.18	0.16	1.00	1.90
Number of observations	4,909			

Appendix A. (cont.)

	Grade 6			
Variable	Mean	Standard deviation	Minimum	Maximum
Math score (absolute)	1133.29	41.05	1012.50	1275.10
Math score (percentile)	52.83	28.32	1.00	100.00
Language score (absolute)	1071.63	33.71	953.90	1197.40
Language score (percentile)	54.04	28.27	1.00	100.00
Class size	26.43	5.77	6.00	38.00
Number of students with similar IQ	5.93	4.13	0.00	22.00
School enrollment	223.40	106.13	23.00	552.00
Weight factor = 1.00	0.53	0.50	0.00	1.00
Weight factor = 1.25	0.37	0.48	0.00	1.00
Weight factor = 1.40	0.00	0.05	0.00	1.00
Weight factor = 1.70	0.00	0.05	0.00	1.00
Weight factor = 1.90	0.09	0.28	0.00	1.00
Gender of student (1 = Female)	0.49	0.50	0.00	1.00
Percent of class female	0.50	0.13	0.00	1.00
Class average weight factor	1.18	0.16	1.00	1.90
Teacher experience	17.65	8.78	1.00	39.00
Dual teacher class	0.38	0.48	0.00	1.00
Gender of teacher (1 = Female)	0.45	0.50	0.00	1.00
Multi-grade class	0.42	0.49	0.00	1.00
Public	0.25	0.43	0.00	1.00
Protestant	0.24	0.43	0.00	1.00
Roman Catholic	0.43	0.50	0.00	1.00
Other	0.08	0.27	0.00	1.00
School average weight factor	1.17	0.16	1.00	1.88
Number of observations	4,574			

Appendix A. (cont.)

	Grade 8			
Variable	Mean	Standard deviation	Minimum	Maximum
Math score (absolute)	1196.55	46.83	1076.40	1361.30
Math score (percentile)	53.48	28.05	1.00	100.00
Language score (absolute)	1118.41	37.11	965.50	1261.20
Language score (percentile)	54.45	27.68	1.00	100.00
Class size	26.82	6.09	5.00	39.00
Number of students with similar IQ	6.56	4.77	0.00	32.00
School enrollment	227.88	117.43	27.00	625.00
Weight factor = 1.00	0.52	0.50	0.00	1.00
Weight factor = 1.25	0.39	0.49	0.00	1.00
Weight factor = 1.40	0.00	0.06	0.00	1.00
Weight factor = 1.70	0.00	0.06	0.00	1.00
Weight factor = 1.90	0.09	0.28	0.00	1.00
Gender of student (1 = Female)	0.51	0.50	0.00	1.00
Percent of class female	0.51	0.13	0.00	1.00
Class average weight factor	1.18	0.16	1.00	1.90
Teacher experience	19.23	7.81	1.00	40.00
Dual teacher class	0.39	0.49	0.00	1.00
Gender of teacher (1 = Female)	0.26	0.44	0.00	1.00
Multi-grade class	0.47	0.50	0.00	1.00
Public	0.26	0.44	0.00	1.00
Protestant	0.24	0.43	0.00	1.00
Roman Catholic	0.41	0.49	0.00	1.00
Other	0.08	0.27	0.00	1.00
School average weight factor	1.18	0.15	1.00	1.90
Number of observations	4,806			